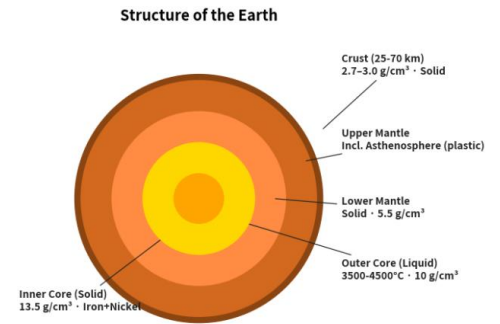


S.3 Geography

Topic 2 – Living with Natural Hazards



REVISION NOTES

Quick-Reference Exam Summary

Topics covered:

1 Structure of the Earth	2 Plate Tectonics & Boundaries
3 Earthquakes (formation & types)	4 Impacts and Measures
5 Level of Destruction	6 Weather, Climate & Cyclone

Colour code key:

Definitions	Remember	Mistakes	Tips	Memory aids	Examples
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Key Terms Glossary

Memorise these definitions — they appear in almost every exam.

- GEN** ▶ **Natural hazard:** An event caused by natural forces that threatens life and property.
- TEC** ▶ **Tectonic hazard:** Hazard caused by movement of Earth's plates (earthquake, volcanic eruption, tsunami).
- TEC** ▶ **Lithosphere:** Crust + uppermost solid part of upper mantle. Broken into tectonic plates.
- TEC** ▶ **Asthenosphere:** Soft, **plastic/molten** part of upper mantle. Plates float and move on it.
- PLT** ▶ **Tectonic plate:** Pieces of lithosphere with **irregular shapes**, different **thickness** and **sizes**; move ~**5–10 cm/year**.
- PLT** ▶ **Plate boundary:** The border between two plates.
- PLT** ▶ **Mantle convection:** Hot rock rises (lower mantle), cool rock sinks (near surface) — drives plate motion.
- EQK** ▶ **Earthquake:** The **sudden shaking** of rocks in the Earth's crust.
- EQK** ▶ **Focus (震源):** The point where **energy is released**. **MUST be underground**.
- EQK** ▶ **Epicentre:** The point on Earth's surface **vertically above the focus**.
- EQK** ▶ **Seismic wave (地震波):** Energy from the earthquake transmitted through the ground in all directions.
- MSR** ▶ **Magnitude:** Amount of energy released from focus. **Richter scale** — **fixed for all places**.
- MSR** ▶ **Intensity:** Degree of damage at a particular place. **Modified Mercalli** — **different at different places**.
- CLM** ▶ **Weather:** Conditions of the lowest atmosphere over a **SHORT** period of time.
- CLM** ▶ **Climate:** Average weather conditions of a place over a **LONG** period (e.g. 30 years).
- CLM** ▶ **Tropical cyclone (typhoon):** An intense **low-pressure** system with closely-packed, circular isobars.

Badges: **GEN**=general **TEC**=tectonics **PLT**=plates **EQK**=earthquake **MSR**=measurement **CLM**=climate

1. Structure of the Earth

3 layers arranged by density, pressure & temperature:

Layer	Sub-layers	State
Crust	Continental (thicker, less dense) Oceanic (thinner, denser)	Solid
Mantle	Upper mantle (incl. asthenosphere) Lower mantle	Solid / plastic/molten (asthenosphere) Solid (lower)
Core	Outer core Inner core	Liquid (outer) Solid (inner)

⚡ Common mistake (MCQ-2 / 2014-Q9)

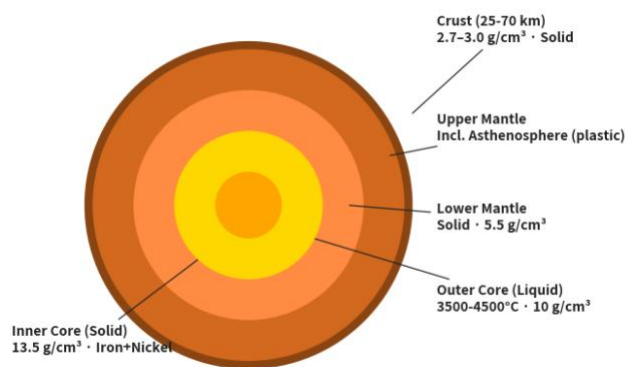
Lower mantle is **SOLID** — not molten. Only the **asthenosphere** (in upper mantle) is plastic/molten.

Outer core is **LESS dense** than inner core (10 vs 13.5 g/cm³).

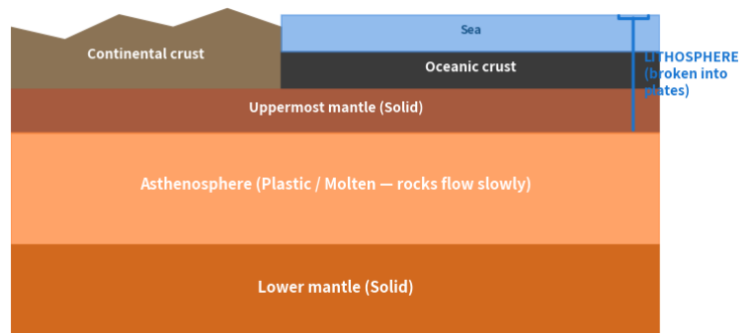
🔗 Lithosphere vs Asthenosphere

- **Lithosphere** = Crust + uppermost SOLID part of upper mantle. Broken into **plates**.
- **Asthenosphere** = soft, plastic part of upper mantle. Plates **float and move on it**.

Structure of the Earth



Lithosphere & Asthenosphere



Comparison between Lithosphere and Asthenosphere

Earth Layer	Sub-layer	Physical State	Lithosphere / Asthenosphere
Crust	Continental crust	Solid	LITHOSPHERE (solid) → broken into <i>plates</i>
	Oceanic crust (sea)	Solid	
Upper Mantle	Uppermost part of the upper mantle	Solid	ASTHENOSPHERE (plastic / molten)
	Lower part of upper mantle <i>(soft rocks that flow slowly)</i>	Plastic / Molten	
Lower Mantle	Lower mantle	Solid	Lower mantle <i>(solid)</i>
Core	Outer core	Liquid	Outer core <i>(liquid)</i>
	Inner core	Solid	Inner core <i>(solid)</i>

 **Key idea:** The lithosphere and asthenosphere are defined by **physical state** (solid vs plastic), not by chemical composition.

LITHOSPHERE

= Crust + **uppermost SOLID** part of upper mantle
 → broken into *tectonic plates*

ASTHENOSPHERE

= the **plastic / molten** part of upper mantle, below the lithosphere
 → plates *float and move* on it

2. Plate Tectonics

Plates — must memorise (name + location + spelling)

#	Plate	Where	#	Plate	Where
1	Eurasian	Europe & Asia	9	Juan de Fuca	NW USA coast
2	Arabian	Arabian Peninsula	10	North American	North America
3	African	Africa	11	Cocos	C. America
4	Indo-Australian	India & Aus.	12	Caribbean	Caribbean Sea
5	Antarctic	Antarctica	13	Nazca	<i>(common in exam)</i>
6	Philippine	Philippine Sea	14	South American	S. America
7	Okhotsk	Sea of Okhotsk	15	Scotia	near Antarctica
8	Pacific	Pacific Ocean			

3 types of plate boundary

	(1) Divergent	(2) Convergent	(3) Transform
Movement	Plates move AWAY	Plates move TOWARDS	Plates slide past
Force	Tensional	Compressional	Shearing / lateral
Also called	Constructive	Destructive	Conservative
Driver	Divergence of convection + ridge push	Convergence of convection + slab pull	Convection currents flow past each other
Example	N. American & Eurasian (Atlantic Ocean)	Philippine & Eurasian Indo-Australian & Eurasian	Pacific & N. American (San Andreas Fault)

Memory aid — Plate boundaries (CDC)

Three types match three forces:

- **C**onvergent ↔ **C**ompressional ↔ Collide ↔ Destructive
- **D**ivergent ↔ Tensional ↔ Depart ↔ Constructive
- **T**ransform ↔ Shearing ↔ Slide past ↔ Conservative

★ Formation of Earthquakes — MEMORISE

(depends on the location of the boundary)

1. **Convergence** of convection current in mantle **drags** the X plate and the Y plate to **collide with each other** along the convergent plate boundary under **compressional force**.
2. **Divergence** of convection current in mantle **pulls** the X plate and the Y plate to **move away from each other** along the divergent plate boundary under **tensional force**.
3. Two convection currents in mantle **flow past each other**, the X plate and the Y plate **slide past each other horizontally** along the **transform** plate boundary under **shearing force**.

Locations of plate boundaries

Type	Examples / Belts
Divergent	Mid-Atlantic, Mid-Indian, Antarctic, Southeast Pacific → Mostly UNDER OCEAN . Volcanoes: Iceland, Easter Island.
Convergent	Alpine-Himalayan Belt (Mediterranean → Himalayas → Indian Ocean) Circum-Pacific Belt = Pacific Ring of Fire 環太平洋火山帶 — Japan, Taiwan, Philippines, Indonesia
Transform	San Andreas Fault — 200 km, California USA. N. American + Pacific plate.

3. Earthquake

Distribution

- Most occur at **convergent + transform** boundaries
- Fewer at **divergent (seabed)**; rare away from boundaries
- Earthquake = tectonic hazard **affecting the largest area of the world**

Formation — 6 steps

#	Step
1	Plate tectonics — plates move
2	Stress applied to rocks; held back by friction
3	Pressure builds up within rocks; stress increases
4	When stress > strength , rock breaks and displaces
5	Energy released, transmits as seismic waves
6	Seismic waves move out in all directions → earthquake formed

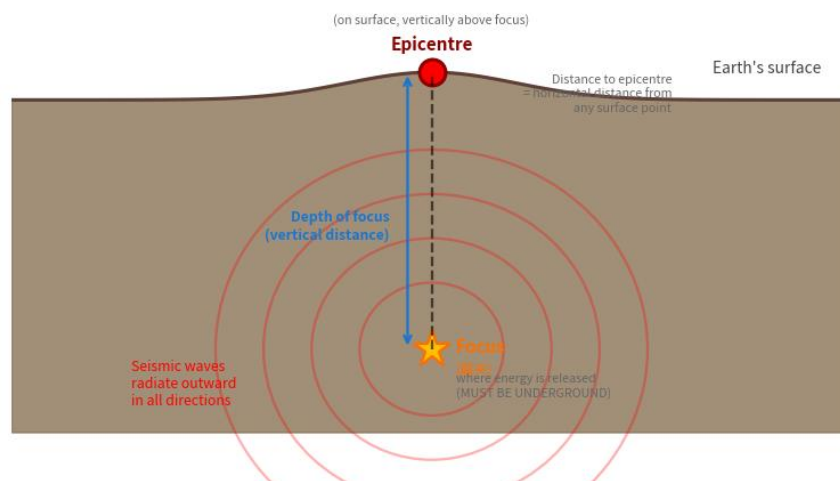
Focus vs Epicentre



Don't confuse these!

- **Focus** = point where energy released — **MUST be UNDERGROUND**.
- **Epicentre** = point on **SURFACE**, vertically above focus.
- Nearer to focus = stronger shaking felt.

Earthquake: Focus and Epicentre



3 ways to describe an earthquake

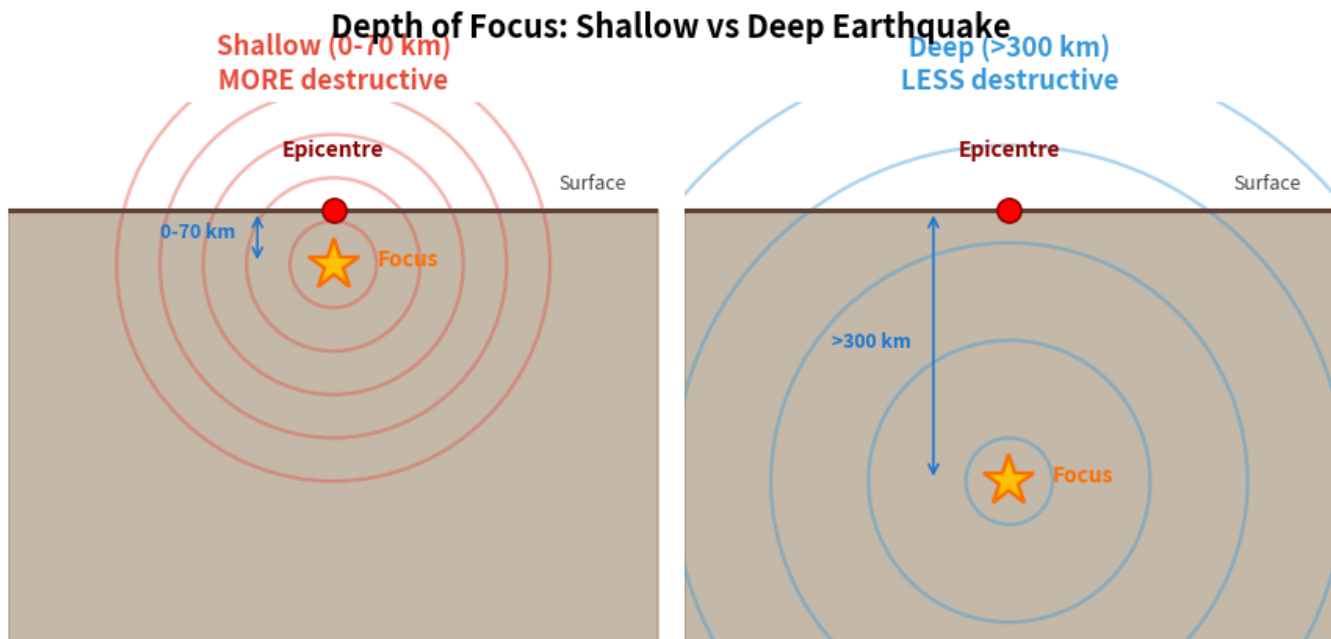
Way	What it measures	Scale	Key point
(1) Depth	Depth in km	Shallow (0-70) Intermediate (70-300) Deep (>300)	Shallow = MORE destructive (less energy loss)
(2) Intensity	Degree of damage at a particular place	Modified Mercalli (I to XII)	Different in different places
(3) Magnitude	Energy released from the focus	Richter scale	Fixed for all places +1 = $\times 31.6$ energy

⚡ Key fact — Richter scale

Each +1 magnitude = $\times 31.6$ more energy released (NOT $\times 10$!)

e.g. M7.0 $\rightarrow$ M8.0 difference = 31.6 times more energy

Depth of Focus: Shallow vs Deep Earthquake



4. Impacts of Earthquake

Short-term effects	Long-term effects
• Destruction of buildings, houses • Damage to transport (roads, rail, bridges) • Hinder rescue work • Destroy public utilities (water/gas/electricity) • Lead to fire (gas leakage) • Tsunami, landslides, flooding • Land subsidence • Spread of disease • Aftershocks (餘震) → Economic loss, loss of life & property	• Obstruction of economic development • Bad psychological effect on victims • Riots & robbery (social instability)

5. Measures — 6 types

#	Measure	Purpose / Key terms
1	Monitoring & warning	Raise awareness; early evacuation. Seismograph, J-Alert (Japan), Shinkansen emergency stop
2	Hazard-proof structures	Earthquake-proof design; laws enforce standard; height controlled
3	Land-use planning	'Hazard-zone map', control development, evacuation routes, open areas as shelters
4	Education & drills	Increase public awareness; emergency kit, education centres, regular drills
5	Rescue systems	Well-trained rescue team, life detectors, helicopters, medicines, wireless networks
6	Post-hazard finance	Rebuild society. Disaster aid, Insurance

6 Measures (M-B-P-E-R-F)

Monitor Build Plan Educate Rescue Fund

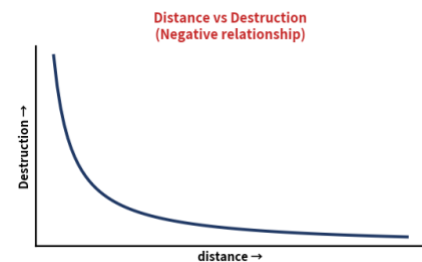
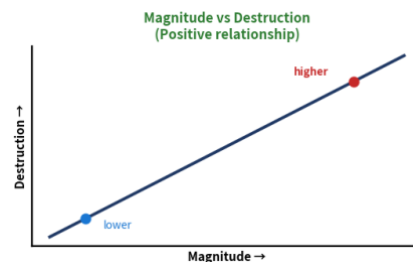
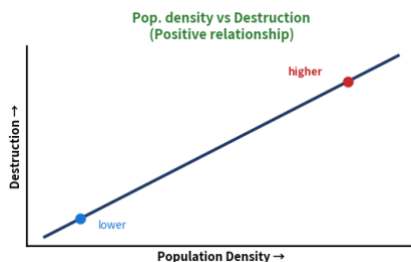
6. Level of Destruction — 3 factor groups

(1) Characteristics of the affected area

Factor	Relationship with destruction
Population density	HIGHER density → more people exposed (positive)
Level of economic dev.	HIGHER dev. → more infrastructure → greater property loss (positive)
Preparedness of people	HIGHER awareness → less loss (negative)
Weather after the hazard	Extreme weather (cold surges, heat waves) increases death toll

(2) Characteristics of the hazard

Factor	Relationship with destruction
Magnitude	Higher magnitude → more energy → more destruction (positive)
Duration	Longer duration → less ability to withstand → more loss (positive)
Time of occurrence	At midnight = sleeping = low awareness → greater destruction
Distance from source	Closer to epicentre (short horizontal) + shallow focus → more destruction (negative)
Frequency	Less frequent hazards → people less prepared
Tsunami wave height	Higher wave → more energy → greater destruction



(3) MDCs vs LDCs — why LDCs suffer more

Factor	Why LDCs suffer more
Economic dev.	Lack capital + technology — can't fund research, warning systems, rescue equipment
Transport network	Poorly developed → rescue work hindered (roads blocked by landslides)
Building strength	Simple materials (stones, bricks, wood); lack earthquake-proof design
Population density	Higher density in LDC cities → higher casualties
Education & awareness	Poor education → less prepared; traditional values resist drills
Government efficiency	Slow response; corruption prevents aid reaching victims; lack of law enforcement on building codes

🧠 Memory aid — LDC suffer more (6 reasons)

Money **R**oads **B**uildings **P**eople (density) **E**ducation **G**overnment

7. Climatic Hazards

Weather vs Climate

	Weather	Climate
Time scale	SHORT period	LONG period (e.g. 30 years)
Definition	Conditions of the lowest atmosphere	Average weather conditions of a place

Elements of weather & their units

Element	Unit	Element	Unit
Wind speed	m/s, km/h (m/s × 3.6 = km/h)	Temperature	°C / °F / K
Wind direction	Compass point (N, NE, E ...)	Relative humidity	%
Rainfall	mm (30 mm/h Amber, 50 mm/h Red)	Air pressure	hPa
Sunshine	hour		

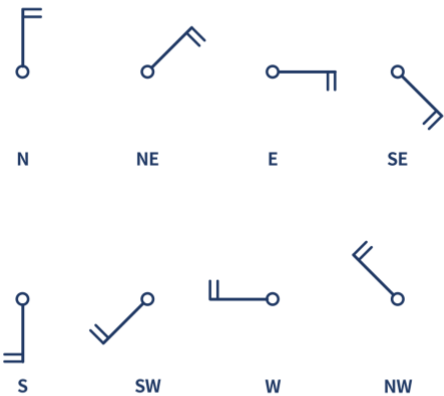
2. Weather Chart (Textbook P.9)

Common Weather Map Symbols (Legends)

Legends

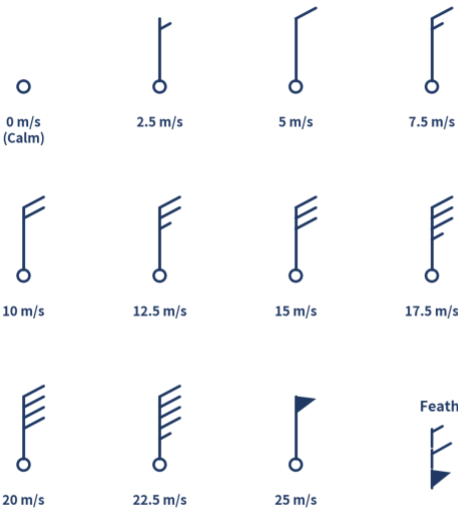
☂ Drizzle	∞ Haze	⎓ 25m/s wind	⚡⚡⚡ Dissipating Cold Front
● Rain	✱ Ice/Snow/mixed with rain	⎓ Isobar (air pressure values in hPa)	— Axis of Trough
▽ Shower	⊞ Sand/Dust	⚡ Cold Front	⊙ Centre of Tropical Cyclone
⚡ Thunderstorm	⊞ Sandstorm/Duststorm	⚡ Warm Front	
≡ Mist	○ Calm or light variable wind	⚡ Occlusion	
≡ Fog	⎓ 5m/s wind	⚡ Stationary Front	

Wind Direction Symbols (8 Compass Directions)



★ The shaft points TOWARDS the direction the wind is coming FROM.
e.g. "N" = wind blowing FROM the north (shaft points up)

Wind Speed Symbols — 0 to 25 m/s



Feather Key
= half feather (2.5 m/s)
= full feather (5 m/s)
= pennant (25 m/s)

★ Always read from TIP of shaft downwards. Build the speed by ADDING feathers.
e.g. 7.5 m/s = 1 full feather (5) + 1 half feather (2.5)

Learn more via Digital Textbook Notes (w/ examples)

Tropical Cyclone (Typhoon / Hurricane)

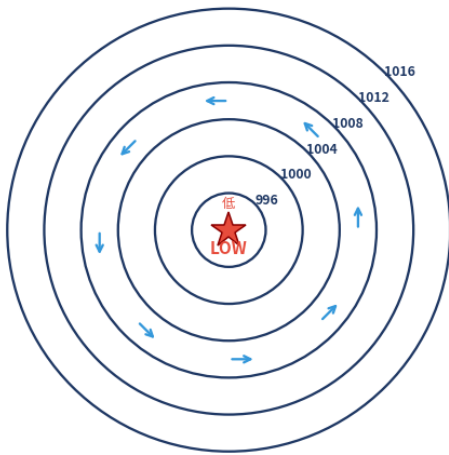
- In HK: occurs **May–November**, most frequent **Aug–Sep**
- Intense **LOW-pressure system**
- Shown by **close, circular isobars** with low-pressure centre
- Strength depends on **maximum sustained wind speed** (最高持續風速)

🌀 Pressure gradient ↔ wind strength

- **Closely-packed** isobars → steep gradient → **STRONG** cyclone (high wind)
- **Widely-spaced** isobars → weak gradient → **WEAK** cyclone (low wind)

HK Tropical Cyclone Classification

Closely-packed isobars → steep pressure gradient → strong cyclone



Tropical Cyclone (Low-Pressure System)

Exam Tips & Common Pitfalls

Always remember

- Focus = **UNDERGROUND** point of energy release (with depth)
- Epicentre = point on **SURFACE** directly above focus (no depth)
- Earthquake = **SUDDEN SHAKING** of rocks in Earth's crust (exact words)
- Magnitude is **FIXED** for all places; Intensity **VARIES** by location
- +1 magnitude = **×31.6 more energy** (NOT ×10)

Common mistakes to avoid

- X Saying lower mantle is molten — only **asthenosphere is plastic**.
- X Confusing focus and epicentre. **Focus = underground; Epicentre = surface**.
- X Saying 'sleeping' instead of '**low awareness**' — use the geography term.
- X Forgetting to say '**horizontal distance**' (to epicentre) vs '**vertical distance**' (depth of focus).

Question-style cheat sheet

- Q:** 'Why is X earthquake-prone?' → Country is near **plate boundary**, convection currents cause plates to collide/diverge/slide → frequent earthquakes.
- Q:** 'Why low death toll despite high magnitude?' → Warnings issued, low pop. density, earthquake-proof buildings, drills.
- Q:** 'Why do LDCs suffer more?' → MRBPEG: Money, Roads, Buildings, People-density, Education, Government.
- Q:** 'Why are tsunamis common here?' → Same as earthquake formation (earthquakes are the trigger).